EE3-07

IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2018

EEE/EIE PART III/IV: MEng, BEng and ACGI

**DIGITAL SIGNAL PROCESSING**

**Corrected copy**

Monday, 10 December 9:00 am

Time allowed: 3:00 hours

**There are FOUR questions on this paper.**

**Answer ALL questions.**

*All questions carry equal marks.*

**Any special instructions for invigilators and information for candidates are on page 1.**

Examiners responsible First Marker(s) : P.A. Naylor
Second Marker(s) : W. Dai

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# DIGITAL SIGNAL PROCESSING

1. a) Compare and contrast FIR and IIR filters. [ 4 ]

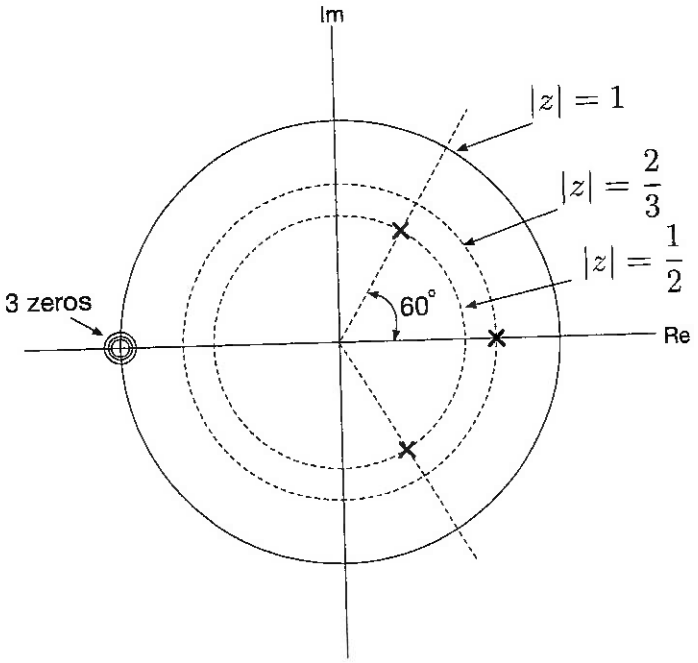



Figure 1.1 Pole-zero diagram

b) The pole-zero plot of a digitial filter is given in Fig. 1.1. The gain of this filter at d.c. is 1.

i) Find the system function of this filter in the form

$$H(z) = A\left[\frac{(1+a_1z^{-1})(1+b_1z^{-1}+b_2z^{-2})}{(1+c_1z^{-1})(1+d_1z^{-1}+d_2z^{-2})}\right]$$

giving the values of the constant coefficients $A$, $a_1$, $b_1$, $b_2$, $c_1$, $d_1$, $d_2$.

[ 8 ]

c) Draw a signal flow graph for the filter of part b) in Direct Form II. [ 4 ]

d) Discuss the impact on the filter's pole-zero plot that would result if the coefficient $d_2$ was found to be erroneously increased by 10%. [ 4 ]


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2. a) Consider the system function given by

$$H(z) = \sum_{k=0}^{\infty} b_k z^{-k}.$$

Describe the characteristics of the filter represented by this system function and state the requirements that must be satisfied for the filter to be causal and stable. [ 4 ]

b) Explain the key properties of a linear phase FIR filter. Include discussion in terms of the filter coefficients and in terms of the z-domain. Include relevant sketches to illustrate your explanation. [ 6 ]

c) Show that the frequency response of a linear phase FIR filter with $N$ coefficients, $N$ even, can be written as

$$H(e^{j\omega}) = e^{j\lambda} \sum_{n=0}^{M-1} 2h(n)\cos(\gamma)$$

and give expressions for $M$, $\lambda$ and $\gamma$. [ 5 ]

d) Draw and label a signal flow diagram of the efficient implementation of a linear phase FIR filter for $N = 9$. [ 5 ]

3. Consider an LTI system operating with sampling frequency 10 kHz and system function

$$H(z) = z^{-1}.$$

a) Draw labelled sketch graphs of the magnitude and phase of the frequency response of this system. [ 2 ]

b) Write down the general expression for group delay and give the units of group delay. Determine the group delay of $H(z)$. [ 3 ]

c) Consider the vector

$$\mathbf{x} = [x(0)\ x(1)\ \cdots\ x(N-1)]^T$$

the elements of which are samples of the input signal and $^T$ indicates the transpose operation. Further consider the vector

$$\mathbf{y} = [y(0)\ y(1)\ \cdots\ y(N-1)]^T$$

the elements of which are samples of the output signal.

i) Given that the processing due to $H(z)$ is formulated as a matrix operation $\mathbf{y} = \mathbf{H}\mathbf{x}$, write out the elements of $\mathbf{H}$. Use the notation $\cdots$ to account for an undefined value for $N$. [ 4 ]

ii) Now write out the elements of $\mathbf{H}_c$ representing the processing due to $H_c(z)$ in matrix form where $H_c(z) = H(z)$ except that any shifts are performed circularly with modulo $N$. [ 3 ]

iii) Given that the DFT is formulated as a matrix operation $\mathbf{y} = \mathbf{F}\mathbf{x}$, write out the elements of $\mathbf{F}$. [ 3 ]

d) Consider writing the processing due to $H_c(z)$ in matrix form so that

$$\mathbf{H}_c = \mathbf{F}^{-1}\mathbf{A}\mathbf{F}.$$

Find the elements of the matrix $\mathbf{A}$. Comment on the significance of this formulation and state any characteristic properties of the rows, columns and diagonals of $\mathbf{F}$ and $\mathbf{A}$ in this context. Discuss briefly the role of normalization in the DFT. [ 5 ]

4. A particular multirate digital signal processing system is formed from an analysis filter bank followed by a synthesis filter bank. The analysis and synthesis filter banks both contain $L$ filters indexed $k = 0, 1, \ldots, L-1$ and are denoted respectively $H_k(z)$ and $G_k(z)$.

a) Draw a block diagram of this multirate system and label the input signal $x(n)$, the output signal $y(n)$ and the filters $H_k(z)$ and $G_k(z)$ for arbitrary $L$. [ 4 ]

b) Discuss the advantages and disadvantages of polyphase implementations of filter banks. [ 2 ]

c) Draw a fully labelled block diagram of the multirate system of a) with $L = 2$ implemented in Type 1 polyphase form, where the polyphase filters are denoted $E_{k\ell}(z)$ for phase $\ell$. [ 3 ]

d) Find the relationship between the coefficients of the filter $E_{k\ell}(z)$ and the coefficients of the filter $H_k(z)$. [ 3 ]

e) Show that the expression for $H_k(z)$ can be written in matrix form as

$$\mathbf{h}(z) = \mathbf{E}(z^L)\mathbf{e}(z)$$

and give the definitions of $\mathbf{h}(z)$, $\mathbf{E}(z)$ and $\mathbf{e}(z)$, where $\mathbf{E}(z)$ is called the polyphase component matrix of the analysis filter bank. [ 4 ]

f) Consider the matrix $\mathbf{R}(z)$ which is the polyphase component matrix of the synthesis filter bank. Define what is meant by *perfect reconstruction* in this context and deduce the condition on $\mathbf{R}(z)$ for perfect reconstruction. [ 4 ]